

Date of publication xxxx 00, 0000, date of current version xxxx 00, 0000.

Digital Object Identifier 10.1109/ACCESS.2024.0000000

Lineage or Era? An Identifiability-Gated Variance Decomposition of Correlated Errors in Public Language Models

S. KANAGA SUBA RAJA¹, SUDHARSAN S¹

¹Department of Computer Science and Engineering, SRM Institute of Science and Technology, Tiruchirappalli, Tamil Nadu 621105, India (e-mail: kanagass@srmist.edu.in; ss0856@srmist.edu.in)

Corresponding author: S. Kanaga Suba Raja (e-mail: kanagass@srmist.edu.in).

This work was supported in part by ...

ABSTRACT Public language models make correlated errors: when one open-weight model fails a question, its contemporaries and descendants often fail it as well. Whether this shared failure structure is driven by lineage—a model replicating the blind spots of the models and data it descends from—or by era—models released together acquiring the same errors through shared training data and technique—is untested, because the two are confounded: descendants are released after their ancestors, making release year both a mediator and a confounder of the lineage path. We introduce an identifiability-gated variance-decomposition instrument, built on the documented release record rather than reconstructed family trees, that partitions correlated error into lineage, era, and model-specific components on the connected subset of the open-model population where the partition is identifiable. The design is gated in two stages. First, a simulation study validates that a crossed restricted-maximum-likelihood estimator recovers known ground truth under realistic occupancy and fails detectably when family and era are nested (aliased). Second, a pre-analysis study-population design procedure—which never observes trait values—selects the smallest population, 22 of 47 models, that remains structurally identifiable and clears a simulation-based recoverability bar. The empirical partition itself is not claimed here: it is the object of a pre-registered measurement pass that this paper scopes, gates, and cost-bounds, and whose lineage-versus-era shares will be reported against a pre-registered decision rule for whether cross-family diversification is a credible remedy for correlated failure.

INDEX TERMS Correlated errors, error correlation, foundation models, identifiability, mixed-effects models, restricted maximum likelihood, variance components, variance decomposition.

I. INTRODUCTION

THE growing reliance on a small number of public open-weight language models raises a statistical problem that is older than the models themselves: when a group of models is given a hard question, they often fail together—missing the same fact, making the same mistake, reaching the same wrong answer. Two explanations compete for this pattern of shared mistakes. The first is *lineage*: a model trained on another model's outputs, or on data generated by it, tends to replicate that model's blind spots. The second is *era*: models released at the same time are trained on largely the same scrapes of the web, the same benchmark-cleansed corpora, and the same dominant techniques, so they may pick up the same errors independently of any shared ancestry [1].

Whether shared failure is mostly inherited or mostly contemporaneous is not an academic question. If shared mistakes

trace mostly to lineage, then diversification across model families is a credible mitigation: deploying models with independent ancestries reduces the probability that the ensemble fails a question the same way. If they trace mostly to era, then no amount of new ancestry helps, because every model of a given year carries the same errors. The choice between these two conclusions has direct operational consequences for how practitioners assemble model portfolios.

The difficulty is that the two explanations are not separable by inspection. Descendants are always released after their ancestors, so release year is simultaneously a *mediator* and a *confounder* of the lineage path: a child model inherits both the errors of its parent and the training environment of its own release date. Any single-number answer that tries to assign shared error “to lineage” or “to era” conflates the two roles of release year.

We address the problem with a variance-decomposition instrument rather than with a causal attribution. Treating the open-model population as a designed experiment, we partition the observed pattern of shared errors into a lineage component σ_L^2 , an era component σ_E^2 , and a model-specific remainder σ_U^2 , estimated with a crossed random-effects (restricted-maximum-likelihood) model. Because the partition is defined only where lineage and era actually vary together—the *connected subset* of the population where the design is crossed rather than nested—the analysis is explicitly scoped to that subset, and an identifiability gate decides whether the decomposition is runnable at all [2], [3].

The central methodological claim of this paper is the ordering of decisions. Identifiability is verified before any trait is measured; the estimator is validated in simulation before any real data are analyzed; and the study population is selected by a pre-analysis design procedure that never observes trait values, before any empirical inference. Section II states the research questions; Section III situates the work; Section IV defines the estimands and identifiability conditions; Section V describes the three stages of the protocol (population audit, simulation validation, pre-analysis population design); Section VI reports the validated results and marks the empirical outputs that are pending the measurement campaign; Sections VII and VIII discuss implications and threats to validity.

II. PROBLEM STATEMENT AND RESEARCH QUESTIONS

A. MAIN QUESTION

The main research question (RQ0) asks whether the observed error-correlation structure across public open-weight language models can be separated into a lineage component and an era component; that is, whether the partition $\sigma_L^2 + \sigma_E^2 + \sigma_U^2$ is estimable and non-degenerate on the connected subset. A positive answer requires both: (a) the estimator survives simulation validation under the realistic population occupancy, and (b) on real data, the lineage and era components are each estimated with non-degenerate intervals.

B. SUB-QUESTIONS

Table 1 lists the sub-questions, their operating population, and their refutation conditions. RQ3 is the primary observational estimand (lineage conditional on era); RQ4 is the mechanistic estimand (lineage with era held fixed), available only on the small co-released and staggered-fine-tune subsets and reported separately. RQ2 is the stop-rule question: a mis-specified generative model must fail *detectably* rather than silently, or the program stops.

C. THE TWO-ESTIMAND RULE

RQ3 and RQ4 answer different questions and are never merged. Release year is both a mediator and a confounder on the lineage-to-error path, so the observational primary adjusts for era grouping but cannot hold era fixed; the mechanistic secondary is the only estimand that isolates lineage with era held exactly fixed, and it exists only where co-released family cohorts and verified cross-generation fine-tune chains

TABLE 1. Research questions and refutation conditions.

RQ	Question	Operates on	Refutation
RQ1	Does the crossed random-effects estimator recover known ground truth under a balanced-crossed (D1) and a realistic-occupancy (D2) regime?	Simulated data	Bias or D2
RQ2	Under a nested (mis-specified) design, does the estimator fail detectably?	Simulated data	Silent bias
RQ3	What share of error covariance is attributable to lineage conditional on era?	Connected subset	$\sigma_L^2 \approx 0$
RQ4	Holding era exactly fixed, what is the structural contribution of lineage?	Co-released cohorts + fine-tune chains	Reported never me
RQ5	After lineage adjustment, does the era component converge across release quarters?	Connected subset	Flat trend
RQ6	Is cross-family diversification a credible mitigation for correlated failure?	Decision layer	Era domi

TABLE 2. Differentiation against closest related work.

Work	Object	What it does
Kim <i>et al.</i> , ICML 2025	agreement rate	measures when models co-er
G-theory / TEE pipelines	pipeline facets	cross-sectional decomposes measurement variance of a pipeline
PhyloLM / lineage reconstruction	phylogeny	reconstructs ancestry trees
Kuai <i>et al.</i> , 2026	behavioral entanglement	independence audit + verification
Monoculture literature	deployed portfolio	reweighting welfare argument given correlated error
This work	error traits on connected subset	identifiability-gated partition $\sigma_L^2/\sigma_E^2/\sigma_U^2$

provide the contrast. We report the two estimands as θ_P (observational) and θ_M (mechanistic), separately, throughout.

III. RELATED WORK

Table 2 positions this work against its closest neighbors. Kim *et al.* measure cross-sectional agreement between language models—how often two models agree when both are wrong—but do not partition the shared variance into components and are explicitly cross-sectional [1]. Generalizability-theory pipelines (TEE-style) apply variance decomposition to measurement pipelines, but to pipeline facets rather than to a population of models whose identifiability must be established first [4]. Phylogeny-reconstruction work treats lineage as a tree to be recovered; we treat it as a variance component of a designed population [5]. The monoculture literature documents the risk of deploying many similar models but treats correlated error as a given input to a welfare argument rather than as an object to be decomposed [6], [7]. Recent work on behavioral entanglement likewise measures how much the behavior of one model can be predicted from another, but stops at an independence audit rather than a trait-level variance partition [8].

No verified prior work estimates $\sigma_L^2 + \sigma_E^2 + \sigma_U^2$ for model error traits, on the provably connected subset, with the estimator’s validity established in simulation first. Every prior approach either documents correlation without partitioning

it, decomposes the variance of a different object, or treats correlated error as a given input to a downstream argument.

IV. FORMAL MODEL AND IDENTIFIABILITY

A. SETUP AND NOTATION

Let the population consist of N open-weight models indexed by $i \in \{1, \dots, N\}$. Each model is released by a family $f(i) \in \{1, \dots, 6\}$ (a verified design grouping, e.g., Llama, Mistral, Qwen, Phi, Gemma, DeepSeek) in a release quarter $e(i) \in \{1, \dots, 14\}$ spanning 2023Q1–2026Q2. Each model receives a continuous per-model trait Y_i (Section V), measured on a fixed common item set. We model the trait as

$$Y_i = \mu + \alpha_{f(i)} + \beta_{e(i)} + u_i, \quad (1)$$

where μ is the grand mean, $\alpha_f \sim \mathcal{N}(0, \sigma_L^2)$ are independent family (lineage) effects, $\beta_e \sim \mathcal{N}(0, \sigma_E^2)$ are independent era effects, and $u_i \sim \mathcal{N}(0, \sigma_U^2)$ are independent model-specific residuals. In vector form,

$$\mathbf{y} = \mu \mathbf{1} + \mathbf{C}\boldsymbol{\gamma} + \mathbf{u}, \quad \boldsymbol{\gamma} \sim \mathcal{N}(\mathbf{0}, \text{diag}(\sigma_L^2, \sigma_E^2)), \quad (2)$$

The causal structure that motivates the two-estimand rule is summarized in Fig. 1: family membership influences the error trait directly and through the release date, and release date is simultaneously a mediator and a confounder of the lineage path, so a single causal attribution is impossible by construction.

where $\mathbf{C}$ is the $N \times 12$ design matrix of family and era indicators with u_i as in (1). The marginal covariance is

$$\mathbf{V}(\theta) = \sigma_U^2 \mathbf{I} + \mathbf{C} \text{diag}(\sigma_L^2, \sigma_E^2) \mathbf{C}^\top, \quad (3)$$

with $\theta = (\sigma_L^2, \sigma_E^2, \sigma_U^2)$.

B. ESTIMATORS

We estimate θ by restricted maximum likelihood (REML), maximizing the log-restricted-likelihood [2], [3]

$$\ell_R(\theta) = -\frac{1}{2} \log \det \mathbf{V} - \frac{1}{2} \log \det (\mathbf{1}^\top \mathbf{V}^{-1} \mathbf{1}) - \frac{1}{2} (\mathbf{y} - \mu \mathbf{1})^\top \mathbf{V}^{-1} (\mathbf{y} - \mu \mathbf{1}), \quad (4)$$

maximized directly with a numerical optimizer. Because $\mathbf{C}$ is low-rank (12 columns), the Woodbury identity

$$\mathbf{V}^{-1} = \sigma_U^{-2} \left(\mathbf{I} - \mathbf{C} (\sigma_U^2 \mathbf{I} + \text{diag}(\sigma_L^2, \sigma_E^2) \mathbf{C}^\top \mathbf{C})^{-1} \mathbf{C}^\top \right) \quad (5)$$

reduces each likelihood evaluation to linear algebra on 12×12 matrices, making repeated simulation feasible. Share confidence intervals use a Monte-Carlo delta method on the log-variances with a numerically differentiated Hessian, PSD-clipped for sparse designs.

We verified this direct implementation against the standard reference implementations. On balanced data the REML estimates reproduce two-way ANOVA method-of-moments estimates exactly (verified at 12×12 , 6×14 , and 8×10 designs). In contrast, the widely used ‘statsmodels’ crossed-variance formulation does *not* maximize the REML objective: on a balanced 12×12 dataset it returns a family/era split away from the optimum (REML objective 66.674

vs. 65.994 for the direct optimizer), understating family-share coverage in simulation. This is an upstream finding worth reporting; all results here use the direct maximizer. The exact comparison—optimizer settings, design, and the two objective values—is reproduced in the released artifact (src/lineage_era/analysis/reml.py) and in the project decision log (2026-08-03).

C. ESTIMANDS

The primary observational estimand is the share vector

$$\theta_P = \left(\frac{\sigma_L^2}{\sigma_L^2 + \sigma_E^2 + \sigma_U^2}, \frac{\sigma_E^2}{\sigma_L^2 + \sigma_E^2 + \sigma_U^2}, \frac{\sigma_U^2}{\sigma_L^2 + \sigma_E^2 + \sigma_U^2} \right). \quad (6)$$

The mechanistic estimand θ_M is the lineage share within designs that hold era exactly fixed: co-released family cohorts and the verified cross-generation fine-tune chains (the only documented parent–offspring edges), reported separately from θ_P .

D. IDENTIFIABILITY CONDITIONS

The partition (6) is estimable only if the design is crossed. We require: (i) every family spans at least two release quarters and at least two quarters contain at least two families; (ii) the induced design matrix has full column rank; (iii) the variance-inflation factors of the design are bounded ($\text{VIF} \leq 10$). A nested design—each family confined to a single era—is perfectly aliased, and the identifiability gate must detect the aliasing rather than silently “succeed” (Section V).

V. METHODOLOGY

The protocol proceeds in three stages, each with an exit gate. The ordering is the contribution: identifiability before results, simulation before real data, and study-population design before empirical inference.

A. PHASE 0: POPULATION AUDIT AND THE CONNECTED SUBSET

We assembled a verified population of $N = 47$ open-weight general models spanning six families and fourteen release quarters (2023Q1–2026Q2). The family $\times$ quarter contingency table was built from Hugging Face API metadata and technical reports, with documented divergences between the public release date and HF timestamps corrected by hand. Two membership rules are central. First, *all claims are scoped to the connected subset*—the maximal subset on which lineage and era vary jointly—so the partition (6) is defined. Second, hosted or closed-weight models are excluded. The resulting design is crossed but unbalanced: 11 of 14 quarters contain at least two families and no family is confined to a single era (Fig. 2).

B. PHASE 1: SIMULATION VALIDATION OF THE ESTIMATOR

Before any real data are analyzed, the estimator is validated on simulated data with known ground truth under three regimes [2]:

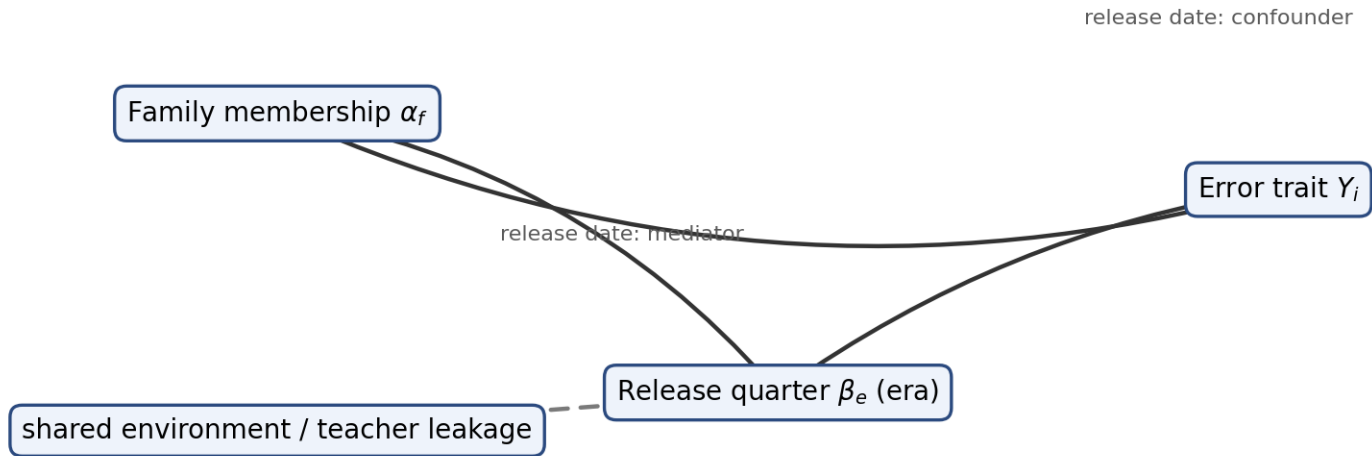


FIGURE 1. Causal structure of the error trait. Family membership and era both influence the error trait; release date is a mediator *and* a confounder of the lineage path, motivating the separate observational (θ_P) and mechanistic (θ_M) estimands. Teacher leakage enters the era (shared environment) channel.

- **D1 (balanced crossed reference).** 30 families $\times$ 14 eras $\times$ 2 models per cell; the large family count isolates estimator calibration from the small-sample limit of the real design.
- **D2 (realistic occupancy).** The occupancy matrix copied from the Phase 0 table (6 families $\times$ 14 quarters, 47 models, sparse cells). Bias or collapse here is a gate failure: the real-data claim never proceeds.
- **D3 (nested; must fail).** Each family confined to a single era, so σ_L^2 and σ_E^2 are perfectly aliased. Three independent detectors (BLUP family-vs-era collinearity, SE/estimate inflation, and profile-likelihood flatness) must flag the aliasing.

A liability decision was resolved in simulation before the trait path was fixed: on item-level binary outcomes at the real occupancy, both a linear-probability model and a binomial GLMM under-estimate the era component and drive it to the boundary in 20–60% of repetitions, whereas continuous per-model traits recover it with 98–100% coverage. Phase 2 therefore uses per-model continuous traits with LPM-REML. A family $\times$ era interaction term was also simulated and documented as non-identified at the sparse occupancy (SE/estimate ratios $\approx 10^4$); the interaction is excluded by design rather than absorbed.

C. PHASE 2: PRE-ANALYSIS STUDY-POPULATION DESIGN (G3)

The empirical campaign measures a fresh trait on all 47 models. Because a full run is expensive, we pre-registered a gate

TABLE 3. G3 optimization inputs.

Input	Used
Family $\times$ quarter occupancy	✓
Lineage graph (verified edges, chain)	✓
Identifiability constraints (rank, VIF, span, crossing)	✓
Cost (public > gated, est. GPU minutes)	✓
Trait values / accuracy	×
Error similarity	×
Evaluation outputs	×

that selects the study population *before any GPU spend*: the smallest connected-subset population whose family $\times$ quarter design is structurally identifiable *and* whose era recovery clears the strict Phase 1 bar in simulation. The key property of this stage is that it is *outcome-independent*: the optimizer’s inputs are occupancy (family $\times$ quarter), the lineage graph, identifiability constraints, and cost—never trait values, accuracies, or evaluation outputs (Table 3).

The procedure is fully pre-registered. It minimizes model count subject to hard constraints (all six families present with at least two quarters of span; every quarter 2023Q1–2026Q2 retains at least one model; the in-subset verified-edge endpoints and the documented chain are retained so the mechanistic estimand survives; the induced design passes the structural checks), then tie-breaks on cost. “Valid,” not merely “identifiable,” requires that the candidate design recover known era shares in a fixed-design DGP battery (scenarios A and B) under the strict bar (absolute era-share bias ≤ 5 pp, CI coverage $\geq 90\%$, convergence $\geq 90\%$). Because

the per-rep share bias is heavy-tailed at this occupancy (standard deviation 22–26pp), a candidate that clears the bar at 300 repetitions must also clear it with a ≥ 1 pp margin at a 1000-repetition confirmation; knife-edge designs sitting exactly on the bar are rejected. Kept models are assigned an auditable reason by single-model ablation—removing the model and asking which design constraint fails—so the population file documents *why* each model is required (era-window coverage, structural identifiability, or statistical recoverability).

D. PHASE 2: TRAIT MEASUREMENT AND DECOMPOSITION

On the selected population, the pipeline is: (i) a fresh five-shot MMLU pass [9] on the common fixed item set (the only compliant source of per-question data; a public reuse audit found no reusable per-question records); (ii) intake validation that aborts on any contract violation of the manifest; (iii) assembly of the continuous per-model trait; (iv) the identifiability gate on the measured design (hard fail = abort); (v) the θ_P partition with REML (4); (vi) bootstrap confidence intervals combined with a trait-error Monte Carlo; and (vii) sensitivity blocks (leave-one-family, leaked-drop, subject-drop, trait-definition, and a documented sanity cross-check against prior leaderboards that is explicitly not a validation). The era-convergence trend and the θ_M mechanistic tables are reported as table entries, separately from the primary partition.

VI. RESULTS

Results status. We report (a) the validated results that exist today—the population audit (Phase 0), the simulation validation (Phase 1), and the pre-analysis population design (G3)—and (b) the empirical outputs of the measurement campaign (Section VI-D), which are pending the pre-registered eval pass and are explicitly marked as such. No empirical claim about lineage-versus-era shares is made in this version; those claims are the object of the pre-registered measurement pass.

A. POPULATION AUDIT (PHASE 0)

The verified population is 6 families $\times$ 14 quarters with $N = 47$ models. The design is crossed (unbalanced/incomplete): 11 of 14 quarters contain at least two families, no family is confined to one era, and 5 cross-generation parent–offspring edges are verified from the metadata (the bulk of true lineage is undocumented and drawn from technical reports). The connected subset—on which all claims are scoped—is the population itself; a nested subpopulation exists and is excluded by the scoping rule. Three caveats are carried into the analysis: the Llama lineage terminates at Llama 4 (post-2025 era variation is carried by the other families); DeepSeek V4 is a ground-up redesign treated as a new independent lineage rather than a descendant; and cross-family teacher leakage (e.g., Phi-4 trained on GPT-4o-generated data) belongs in the era channel as shared environment. Fig. 2 shows the occupancy matrix.

TABLE 4. Simulation validation (Phase 1).

Regime	Recovery / detection	Gate	Verdict
D1 balanced-crossed	share bias ≤ 2.5 pp, coverage 95–96%	PASS	GO
D2 realistic occupancy	share bias ≤ 5.3 pp, coverage 95–100%	PASS*	GO (documented)
D3 nested (aliased)	detection 100%, silent coverage 0%	PASS	GO
Liability (binary)	era underpowered at real occupancy; continuous recovers non-identified (SE ratio $\approx 10^4$)	n/a	path decided
L $\times$ E interaction		n/a	documented

*Family-share bias of -5.3 pp in the lineage-dominant scenario is the six-family small-sample limit, disclosed in Section VIII.

B. SIMULATION VALIDATION (PHASE 1)

Table 4 summarizes the validation. Under the balanced reference (D1, 300 repetitions) share bias is ≤ 2.5 pp with coverage 95–96%. Under the real occupancy (D2) share bias is ≤ 5.3 pp with coverage 95–100%; the family-share bias of -5.3 pp in the lineage-dominant scenario is the documented small-sample limit of six family levels ($df = 5$) and is reported as a limit, not corrected away. The nested design (D3) is detected in 100% of repetitions by all three independent detectors, with zero silent coverage. The liability test resolved the trait path to per-model continuous traits; the family $\times$ era interaction is documented as non-identified rather than estimated. The verdict is GO WITH CHANGES, with the changes being the continuous-trait path and the disclosure of the family-share limit.

C. PRE-ANALYSIS POPULATION DESIGN (G3)

Fig. 3 and Table 5 report the gate. The structural minimum is 21 models (identifiable: full rank, $VIF \leq 10$), but the 21-model design sits exactly on the strict bar in simulation: its era-share bias at 1000–2000 repetitions is ≈ -5.0 pp with a standard error of 0.6–0.8pp—a knife-edge whose verdict flips with the repetition count—so it fails the margin confirmation. The minimum *valid* population is 22 of 47 models, which clears the bar at 300 repetitions (era bias A 2.4pp / B -0.8 pp), passes the 1000-repetition margin confirmation (A 2.2pp / B -2.4 pp), and is robust at 2000 repetitions (A 1.7pp / B -3.2 pp). The full 47-model design passes by the same criterion (A 1.5pp / B -2.2 pp at 1000 repetitions), confirming that the reduction is statistically justified: 22 models recover era shares in the same order of magnitude as all 47, at roughly 67% of the estimated single-GPU cost. The winner keeps all six families, all fourteen quarters, the verified-edge endpoints, and the chain needed for the mechanistic estimand. The population file assigns each of the 22 kept models an auditable reason by single-model ablation (era-window coverage, structural identifiability, or statistical recoverability) and each dropped model the reason “redundant in-cell replication.”

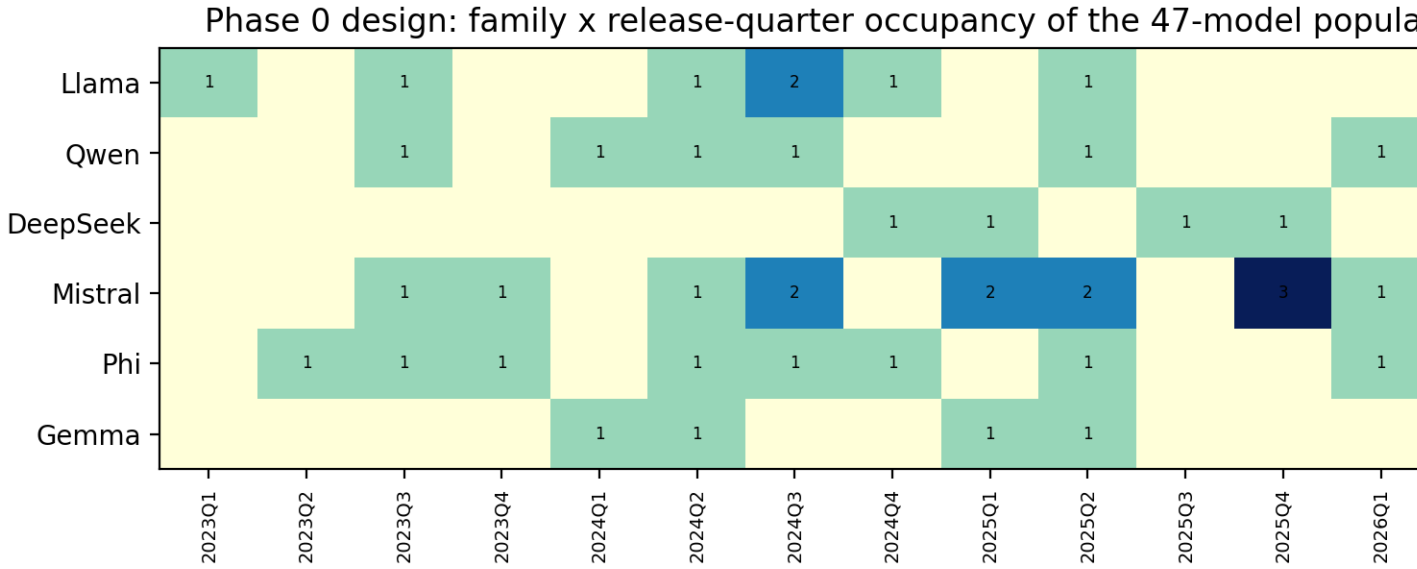


FIGURE 2. Phase 0 design: family $\times$ release-quarter occupancy of the 47-model connected subset. 11 of 14 quarters contain at least two families and no family is confined to a single era; sparse cells carry the small-sample limits reported in Section VIII.

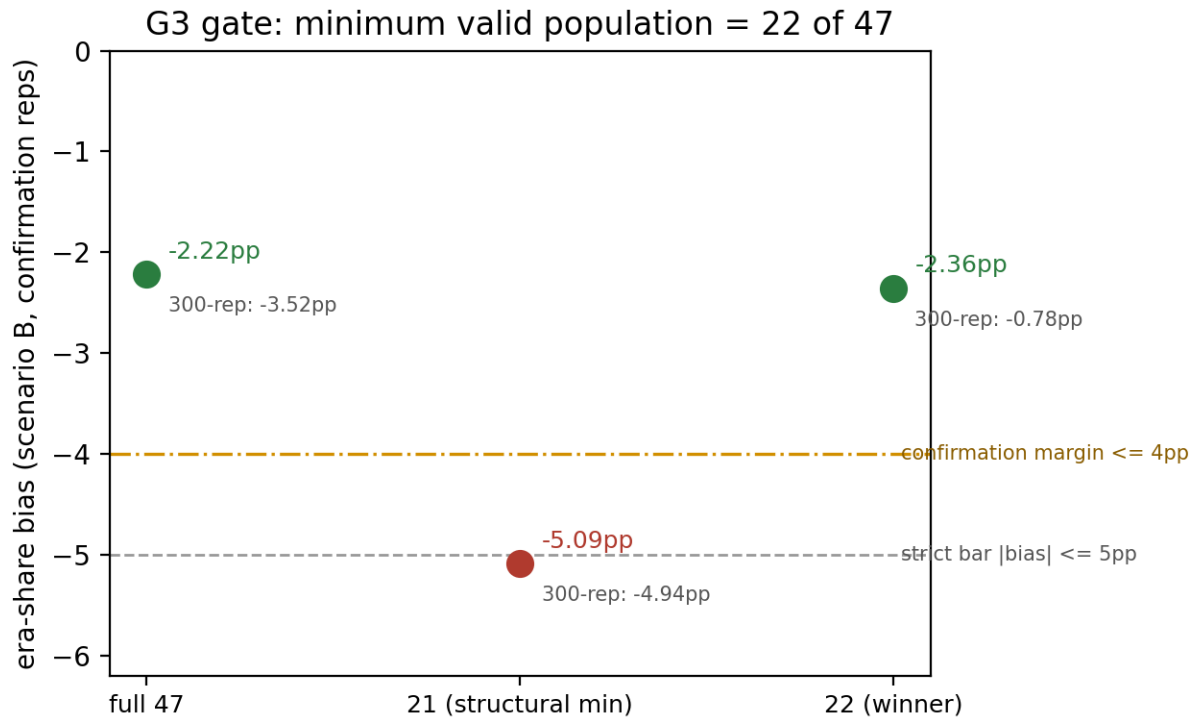


FIGURE 3. G3 decision trace: population size vs. era-share bias (scenario B) with confirmation values. 47 passes, the structural minimum 21 fails the margin confirmation (knife-edge), and 22 passes – the extra model over the structural minimum is statistically necessary, not computationally convenient.

D. EMPIRICAL OUTPUTS (PENDING MEASUREMENT)

The following results are produced by the Phase 2 pipeline on the measured trait and are *pending the fresh eval pass*; they

are listed with the exact outputs that will fill them.

- **Error-similarity panel (supporting).** Observed pairwise error overlap across models, chance-corrected with

TABLE 5. G3 gate: minimum valid population = 22 of 47.

n	models	bias A (pp)	bias B (pp)	cov% (A/B)	confirm A/B	pass
47	47	0.3	-3.5	99 / 96	1.5 / -2.2	True
21	21	1.8	-4.9	96 / 98	2.1 / -5.1	False
22	22	2.4	-0.8	98 / 99	2.2 / -2.4	True

a pre-registered measure (ϕ is locked by the pre-registration rule), with the null ladder (observed $\rightarrow$ matched-accuracy $\rightarrow$ item-difficulty $\rightarrow$ independence). Outputs: overlap heatmap, network, and embedding (Fig. 4).

- **θ_P variance partition.** REML estimates and bootstrap intervals for σ_L^2 , σ_E^2 , σ_U^2 on the 22-model population (Fig. 5). (TBD).
- **Identifiability audit on the measured design.** Rank, VIF, and the D3-must-fail detectors re-run on the real occupancy. (TBD).
- **Era-convergence entry.** Era share by release quarter after lineage adjustment; reported as a table entry. (TBD).
- **θ_M mechanistic tables.** Family contrasts within co-released quarters and per-quarter slopes along the verified fine-tune edges, reported separately. (TBD).
- **Sensitivity blocks.** Leave-one-family, leaked-drop, subject-drop, trait-definition, and the documented leaderboard sanity check. (TBD).

VII. DISCUSSION AND INTERVENTION IMPLICATIONS

The partition θ_P distinguishes two actionable regimes. If the lineage share dominates, cross-family diversification is a credible mitigation: models with independent ancestries add error variance that is not shared, so an ensemble is less likely to fail a question the way its members do. If the era share dominates, diversification alone is insufficient, because models released in the same period share the same training environment regardless of ancestry. The decision rule that follows from RQ6 is intentionally conservative: only a dominant era share refutes diversification-as-remedy for the connected population; a dominant lineage share is a necessary, not sufficient, condition for recommending diversification, because the connected subset may not represent the broader model universe. Independent recent work on ensemble combination reaches the same boundary from the other side: pairwise error correlation substantially *underestimates* the co-failure ceiling of an ensemble, so the operative risk is the joint all-wrong rate [10]; our partition estimates exactly the lineage and era shares whose sum governs that ceiling on the connected population.

Two structural safeguards keep the discussion honest. First, the mechanistic estimand θ_M is reported separately and never merged into θ_P ; only the small co-released cohorts and staggered fine-tune chains can hold era exactly fixed, so any mechanistic claim is scoped to them. Second, all conclusions are scoped to the connected subset of the open-model population as defined in Section V-A; the partition is undefined by construction on nested subpopulations, which is a property of the design, not a data gap.

VIII. THREATS TO VALIDITY AND LIMITATIONS

- **Six-family small-sample limit.** The real design has six family levels ($df=5$), which caps family-share CI coverage below nominal regardless of CI construction and produces a family-share bias of order $-5pp$ in the lineage-dominant scenario (F2 in the simulation report). Family shares are reported with this limit disclosed; the same limit argues for acquiring more lineage families or reporting family shares as lower bounds.
- **Lineage metadata sparsity.** True cross-generation lineage is largely undocumented in model cards; only five cross-generation edges are verified. This constrains the mechanistic estimand, not the primary partition, which uses the verified family grouping.
- **Teacher leakage.** Cross-family teacher-student relationships (e.g., Phi-4 from GPT-4o data, Gemma from Gemini) violate the independence assumption to a known-but-unknown degree and are disclosed as belonging to the era channel.
- **Underpowered binary era.** Item-level binary outcomes cannot resolve the era variance at this occupancy; the continuous per-model trait path is the documented resolution, and any binary-era claim must carry the underpower caveat.
- **Independent trait-error.** Per-model trait measurement errors are assumed independent across models; correlated measurement error would move shares jointly, and is probed by the trait-definition and leaked-drop sensitivity blocks.
- **Common item set.** A fresh five-shot pass on one fixed item set is the basis for all traits; prior work mixing leaderboard subsets is explicitly not used as validation.

IX. CONCLUSION

We have presented an identifiability-gated variance-decomposition instrument for correlated errors in public language models, and the decision order that makes it defensible: verify identifiability, validate the estimator in simulation, design the study population without observing outcomes, and only then decompose measured traits into lineage, era, and model-specific components on the connected subset. The estimator is validated under both balanced and realistic occupancy and fails detectably under nested mis-specification. A pre-analysis procedure selects a minimum valid population of 22 of 47 models—the smallest population that is structurally identifiable and clears the simulation-based recoverability bar—at about 67% of the full-run cost. The empirical partition—the actual lineage and era shares on the selected population—is the outcome of the pre-registered measurement pass and is deliberately not anticipated here; the decision rule for whether cross-family diversification is a credible mitigation for correlated failure is pre-registered and will be evaluated from those shares. All artifacts—population, design, the optimizer, the validator, and the tests—are released with the paper.

Error-overlap panel — pending the pre-registered eval pass

ise phi (chance-corrected) + null ladder: observed $\rightarrow$ matched-accuracy $\rightarrow$ item-difficulty $\rightarrow$ independ

FIGURE 4. Observed pairwise error overlap between models, chance-corrected (pre-registered phi measure) with the null ladder. Pending the fresh eval pass.

θ_P variance partition — pending the fresh eval pass

σ_L^2 (lineage) / σ_E^2 (era) / σ_U^2 (model-unique), REML + bootstrap, 22-model connected subset

FIGURE 5. θ_P variance partition: lineage, era, and model-unique shares with bootstrap intervals on the 22-model connected subset. Pending the fresh eval pass.

APPENDIX A PHASE 0 VERIFICATION

Family groupings were verified against Hugging Face organization metadata and technical reports. Release quarter is the public release date, not the HF *createdAt* timestamp; four documented divergences were corrected by hand. DeepSeek V4 is treated as a new independent lineage (a ground-up

redesign that dropped the MLA architecture), and the within-family Small chain (Small 3 $\rightarrow$ 3.1 $\rightarrow$ 3.2 $\rightarrow$ 4; Devstral-2 on Small-3.1-Base) is the only verified cross-generation chain.

APPENDIX B DETECTABLE-FAILURE BATTERY (D3)

The nested mis-specification is flagged by three independent detectors: (i) BLUP family-vs-era collinearity; (ii)

SE/estimate inflation; and (iii) profile-likelihood flatness along the aliased direction. A design passes the gate only if all three flag the aliasing in $\geq 90\%$ of repetitions and no silent CI coverage is observed.

APPENDIX C COMPUTATION

All Phase 0 and Phase 1 analyses run on CPU. The Phase 2 eval pass runs on one $8 \times H200$ -141GB node in fp8; per-model memory and GPU-time plans are released with the artifacts. The direct REML maximizer reduces each likelihood evaluation to 12×12 linear algebra via the Woodbury identity (5). The estimator comparison and all reported statistics are reproducible from the released code and the population/design files; see the artifact manifest in the repository root for the file-by-file map.

ACKNOWLEDGMENT

REFERENCES

- [1] E. Kim, A. Garg, K. Peng, and N. Garg, "Correlated errors in large language models," in *Proc. Int. Conf. Mach. Learn. (ICML)*, vol. 267, pp. 30038–30066, 2025. [Online]. Available: <https://arxiv.org/abs/2506.07962>
- [2] D. A. Harville, "Maximum likelihood approaches to variance component estimation and to related problems," *J. Amer. Statist. Assoc.*, vol. 72, no. 358, pp. 320–338, 1977.
- [3] H. D. Patterson and R. Thompson, "Recovery of inter-block information when block sizes are unequal," *Biometrika*, vol. 58, no. 3, pp. 545–554, 1971.
- [4] R. L. Brennan, *Generalizability Theory*. New York, NY, USA: Springer, 2001.
- [5] N. Yax, P.-Y. Oudeyer, and S. Palminteri, "PhyloLM: Inferring the phylogeny of large language models and predicting their performances in benchmarks," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2025. [Online]. Available: <https://arxiv.org/abs/2404.04671>
- [6] B. Hedden and M. Raghavan, "Algorithmic monoculture and its critics," 2026. [Online]. Available: <https://arxiv.org/abs/2604.06047>
- [7] N. Jo, N. Garg, and M. Raghavan, "The subjectivity of monoculture," 2026. [Online]. Available: <https://arxiv.org/abs/2602.24086>
- [8] C. Kuai *et al.*, "How independent are large language models? A statistical framework for auditing behavioral entanglement and reweighting verifier ensembles," 2026. [Online]. Available: <https://arxiv.org/abs/2604.07650>
- [9] D. Hendrycks *et al.*, "Measuring massive multitask language understanding," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021.
- [10] J. Chen, "When does combining language models help? A co-failure ceiling on routing, voting, and mixture-of-agents across 67 frontier models," 2026. [Online]. Available: <https://arxiv.org/abs/2606.27288>

S. KANAGA SUBA RAJA received the B.E. degree in Computer Science and Engineering and the M.E. degree in Computer Science and Engineering, and is with the Department of Computer Science and Engineering, SRM Institute of Science and Technology, Tiruchirappalli. His research interests include ...

SUDHARSAN S is with the Department of Computer Science and Engineering, SRM Institute of Science and Technology, Tiruchirappalli. His research interests include ...

...